e otherwise,

where:

min is 7° for passenger ships and 25° for cargo ships; and

max is 15° for passenger ships and 30° for cargo ships.

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4 The factor smom,i is applicable only to passe nger ships (for cargo ships smom,i shall
be taken as unity) and shall be calculated at the final equilibrium from the formula:

()
heel
max
mom,
M
ntDisplaceme04 . 0 GZs i =
where:

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SOURCE: MSC.194(80).pdf
4 The factor smom,i is applicable only to passe nger ships (for cargo ships smom,i shall
be taken as unity) and shall be calculated at the final equilibrium from the formula:

()
heel
max
mom,
M
ntDisplaceme04 . 0 GZs i =
where:

Displacement is the intact displacement at the subdivision draught;

Mheel is the maximum assumed heeling moment as calculated in accordance with
paragraph 4.1; and

smom,i 1

4.1 The heeling moment Mheel is to be calculated as follows:

Mheel = maximum { Mpassenger or Mwind or MSurvivalcraft}

4.1.1 Mpassenger is the maximum assumed heeling moment resulting from movement of
passengers, and is to be obtained as follows:

Mpassenger = (0.075 Np) (0.45 B) (tm)

where:

Np is the maximum number of passengers pe rmitted to be on board in the service condition corresponding to the deepest subdivision draught under consideration; and

B is the beam of the ship.

Alternatively, the heeling moment may be calculated assuming the passengers are distributed with 4 persons per square metre on available deck areas towards one side of

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SOURCE: MSC.194(80).pdf
and

B is the beam of the ship.

Alternatively, the heeling moment may be calculated assuming the passengers are distributed with 4 persons per square metre on available deck areas towards one side of the ship on the decks where muster stations are located and in such a way that they produce the most adverse heeling moment. In doing so, a weight of 75 kg per passenger is to be assumed.

4.1.2 Mwind is the maximum assumed wind force acting in a damage situation:

Mwind = (P A Z) / 9,806 (tm)

where:

P = 120 N/m 2;

A = projected lateral area above waterline;

Z = distance from centre of lateral projected area above waterline to T/2; and

T = ship’s draught, di.

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4.1.3 MSurvivalcraft is the maximum assumed heeling moment due to the launching of all fully loaded davit-launched survival craft on one side of the ship. It shall be calculated using the following assumptions:

.1 all lifeboats and rescue boats fitted on the side to which the ship has heeled

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SOURCE: MSC.194(80).pdf
fully loaded davit-launched survival craft on one side of the ship. It shall be calculated using the following assumptions:

.1 all lifeboats and rescue boats fitted on the side to which the ship has heeled after having sustained damage sha ll be assumed to be swung out fully loaded and ready for lowering;

.2 for lifeboats which are arranged to be launched fully loaded from the stowed position, the maximum heeli ng moment during launching shall be taken;

.3 a fully loaded davit-launched liferaft attached to each davit on the side to which the ship has heeled after having sustained damage shall be assumed to be swung out ready for lowering;

.4 persons not in the life-saving appl iances which are swung out shall not provide either additional heeling or righting moment; and

.5 life-saving appliances on the side of the ship opposite to the side to which the ship has heeled shall be assumed to be in a stowed position.

5 Unsymmetrical flooding is to be kept to a minimum cons istent with the efficient arrangements. Where it is necessary to correct large angles of heel, the means adopted shall,

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the ship has heeled shall be assumed to be in a stowed position.

5 Unsymmetrical flooding is to be kept to a minimum cons istent with the efficient arrangements. Where it is necessary to correct large angles of heel, the means adopted shall, where practicable, be self-acting, but in a ny case where controls to equalization devices are provided they shall be operable from above the bulkhead deck. These fittings together with their controls shall be acceptable to the Administration. Suitable information concerning the use of equalization devices shall be supplied to the master of the ship.

5.1 Tanks and compartments taking part in such equalization shall be fitted with air pipes or equivalent means of sufficient cross-se ction to ensure that the flow of water into the equalization compartments is not delayed.

5.2 In all cases, si is to be taken as zero in those cases where the final waterline, taking into account sinkage, heel and trim, immerses:

.1 the lower edge of openings through which progressive flooding may take place and such flooding is not accounted for in the calculation of factor si.

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SOURCE: MSC.194(80).pdf
into account sinkage, heel and trim, immerses:

.1 the lower edge of openings through which progressive flooding may take place and such flooding is not accounted for in the calculation of factor si.

Such openings shall include air-pipes, ventilators and openings which are closed by means of weathertight doors or hatch covers; and

.2 any part of the bulkhead deck in passenger ships considered a horizontal evacuation route for compliance with chapter II-2.

5.3 The factor si is to be taken as zero if, taking into account sinkage, heel and trim, any of the following occur in any intermediate stage or in the final stage of flooding:

.1 immersion of any vertic al escape hatch in the bulkhead deck intended for compliance with chapter II-2;

.2 any controls intended for the opera tion of watertight doors, equalization devices, valves on piping or on ventilation ducts intended to maintain the integrity of watertight bulkheads fr om above the bulkhead deck become inaccessible or inoperable;
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inaccessible or inoperable;
RESOLUTION MSC.194(80)